

Adaptive Retrieval-Augmented Generative Intelligent Tutoring System for Automata Theory

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Abstract— Over the past few years, large language models (LLMs) and retrieval-augmented generation (RAG) have redefined what is possible in various fields, and the field of AI-driven education is no different. Studies have shown that fewer students have been reading books related to their courses, instead heavily relying on lecture notes offered by their professors, which ultimately doesn't always suffice, especially for core, foundational courses. Automata Theory, a course fundamental for Computer Science and Engineering majors, is a subclass of theoretical computer science that focuses on abstract machines (automata) and the formal languages they recognize. Through this paper, we detail the architecture and preliminary assessment of a customized intelligent tutoring system that provides a comprehensive and adaptive learning environment for a formally rigorous subject like Automata Theory. Our primary contribution lies in the synergistic integration of a specialized pipeline that addresses key limitations of standard RAG approaches. This includes: a hierarchical segmentation method that preserves the pedagogical structure of the textbook, the generation of context-aware summaries and quizzes, and a fully integrated, RAG-powered chatbot specialized in Automata Theory. The system personalizes learning paths and is specifically designed to handle the scale and complexity of a full-course textbook, aiming to help students study the course more effectively. Preliminary evaluations from a pilot study have shown highly positive user reception, with participants self-reporting enhanced comprehension and a significant increase in their motivation to engage with the subject matter.

Keywords—*Intelligent Tutoring Systems, Automata Theory, Retrieval-Augmented Generation (RAG), Large Language Models, Conversational AI, Vector-based Semantic Search (FAISS), Educational Technology*

I. INTRODUCTION

The field of education has long stood at the heart of the progress of societies, empowering individuals to think critically, solve complex problems, and innovate for the advancement of humanity. Since the 1960s, Artificial Intelligence (AI) has been an integral part of the education field [1]. A prominent example of the application of AI in education is the development of intelligent tutoring systems (ITS). These computer-based programs function as personalized tutors, offering adaptive instructions and personalized feedback. Distinguished studies have demonstrated that compared to traditional learning methods, studying through an ITS can raise test scores by 0.66 standard deviation scores, equivalent to a shift from the 50th to the 75th percentile [2]. Remarkably, recent significant breakthroughs in AI and Machine Learning (ML) are transforming the educational landscape, highlighting the promising potential of an ITS.

Online learning platforms have increasingly incorporated machine-learning components such as deep learning, natural language processing, and reinforcement learning [3]. However, the same systematic review highlights an absence

of fully integrated AI tutors that embed these techniques directly into authentic courseware. The subject of Automata Theory presents a significant challenge for intelligent tutoring, as its formally rigorous and conceptually dense content is difficult for students to master with traditional, passive resources like textbooks and lecture notes alone. The effort to address these challenges with technology is well-established. For instance, a seminal work [4], demonstrated over a decade ago that an integrated virtual environment with simulators and visual examples could significantly improve a student's performance and understanding of the subject. Our research builds upon this pedagogical foundation but capitalizes on the power of modern Generative AI to create a system that is not only interactive but also deeply adaptive and conversational in ways not possible with earlier technologies. Our ITS is built to bridge this pedagogical gap through a novel architecture that integrates multiple learning modalities into a single, unified ecosystem. This architecture features context-aware content generation, capitalizing on a unique hierarchical segmentation method to produce coherent summaries and quizzes, an interactive module for active skill development where students construct Finite State Machines and receive immediate AI-driven feedback, a support system with a RAG-powered chatbot, and personalized progress tracking all seamlessly integrated into a single system. The entire learning environment is powered by a resilient backend utilizing FAISS [5] for efficient semantic retrieval and the Gemini 2.0 Flash model for sophisticated content generation and analysis.

This paper begins by reviewing related work in ITS and RAG methods (Section II), followed by a detailed discussion of our system design and implementation (Section III). The next section presents an evaluation of our ITS's effectiveness based on a pilot study involving undergraduate students who have previously completed the Automata Theory course at our university (Section IV). The results have demonstrated encouraging preliminary signs and outcomes. Finally, this paper concludes with lessons learned and outline future directions for deploying AI-driven tutors such as the one we have developed (Section V).

II. RELATED WORK

RAG combined with GenAI models has paved the way for numerous powerful ITS and agents capable of significantly improving students' learning experiences and expanding their knowledge [6], [7]. This section examines two prominent examples.

A. QMS-course RAG agent

Ye et al. [8] propose an innovative use case of RAG and GenAI, like our system. For instance, both platforms offer conversational means of support that grounds a LLM's output from provided course materials by embedding the material into a vector store, retrieving the top-k relevant chunks for

each student query, appending them to the prompt, and outputting a response. However, some notable differences can also be observed.

A key distinction lies in our content processing pipeline. Ye et al. [8] employ a general, flat segmentation strategy for their course materials. Such methods have demonstrated significant weaknesses with large, structured documents, often leading to context loss. Our novel hierarchical segmentation method is a direct solution to this problem, preserving the book's intrinsic chapter/subchapter structure to ensure contextual integrity for both summarization and retrieval.

Furthermore, [8] is exclusively an agent. A stateless RAG component that retrieves texts or practice items on demand from the Quality Management System materials, without tracking any user progress or providing interactive user interfaces. In contrast, our ITS is a fully interactive system that automatically tracks and analyzes each student's progress, providing deeper engagement and analysis.

B. Iris

A highly relevant contribution in this area, Iris is an AI-driven virtual tutor for computer science programming exercises developed by Bassner et al. [9]. Integrated into the Artemis learning platform, Iris is designed around the pedagogical principle of "calibrated assistance," aiming to foster independent problem-solving rather than providing direct solutions. It uses a Chain-of-Thought prompting strategy with GPT-3.5-Turbo to offer students subtle hints or counter-questions. The system achieves its contextual awareness by accessing the problem statement, the student's code, and automated test feedback from the platform. The authors' empirical evaluation found that students perceive Iris as an effective and non-judgmental learning aid, viewing it as a valuable complement to human tutors, particularly for asking questions they might otherwise hesitate to voice.

The Iris system shares a core philosophy with our ITS, as both are engineered to guide students through a dialectical interaction instead of simply providing answers. Our work, however, builds upon and differs from this approach in two critical aspects. Firstly, our ITS is specifically designed for the formally rigorous and abstract domain of Automata Theory, featuring specialized tools like an interactive FSM drawing module that addresses the subject's unique visual and conceptual challenges. Secondly, and most significantly, our system's core architecture realizes the very "future work" proposed by the authors of Iris. They identify their method of context retrieval (selecting from a list of entire files) as a limitation and suggest that an ideal solution would involve "building an embedding index of the files" for more precise context. Our system's Retrieval-Augmented Generation (RAG) pipeline, built on a FAISS vector index, is a direct implementation of this advanced approach, allowing our ITS to ground its responses in the most relevant, granular segments of the source material, thereby offering a more focused and scalable solution for complex academic texts.

C. GPTutor

Lui et al.'s GPTutor [10] represents a prominent use case of a GenAI-powered ITS. GPTutor employs a RAG pipeline over instructor-provided lecture materials to deliver interactive, knowledge-grounded question answering, summaries, and formative quizzes. Although similar in many aspects, one key advantage GPTutor possesses is that it allows the instructor to view and monitor their students' progress, an important feature that we aim to add in the future. However, GPTutor exhibits some key limitations including repetitive content generation, a lack of visual support, and limited course progress tracking.

When generating study content through the uploaded lecture materials, Lui et al. observed that GPTutor frequently repeated content from the lecture material rather than offering new insights. Moreover, students expressed a desire for more in-depth explanations that extend beyond the original content [10]. They also suggested generating more advanced content from the lecture notes, "such as diagrams and tables, to enhance engagement" [10]. Our ITS offers solutions to these shortcomings by generating concise summaries distilled from the textbook PDF and paired with formative quizzes designed to promote deeper understanding, rather than just echoing source text. Additionally, our ITS natively extracts and displays figures from the provided Automata Theory textbook as shown in Fig. 1.

Another notable shortcoming is GPTutor's lack of comprehensive progress tracking. While GPTutor provides students with a dashboard displaying the overall engagement, users wished for detailed indicators, specifically "milestones to track study progress" [10]. Our ITS addresses this by automatically logging the student's chapters read, quizzes attempted, and modules completed (both quizzes and chapter summaries), thereby enabling personalized review paths.

By addressing these limitations through automated summarization, figure extraction, rigorous RAG grounding, and comprehensive progress logging, our ITS advances the practical deployment of AI tutors in formal STEM domains.

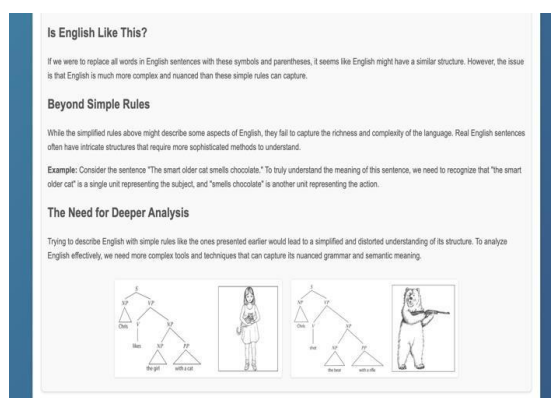


Fig. 1. Screenshot showing our system's ability to display figures and diagrams within the generated summarized content.

III. SYSTEM ARCHITECTURE AND COMPONENTS

An ITS is composed of four integral modules: an expert module, a tutor module, a student diagnosis module, and a user interface module [11]. In our system, the expert module is resembled in the textbook pdf of the Automata Theory book ingested. The contents of the pdf were split into overlapping chunks, and a FAISS index was built to serve as the knowledge base for the system. The tutor module is represented by the RAG pipeline retrieving the top-k relevant chunks, then uses Gemini 2.0 Flash carefully prompted templates to produce well-rounded relevant answers, concise summaries, and formative quizzes. The student diagnosis module is represented by the real time tracking of each student's progress, recording their data and saving them to Firebase's Cloud Firestore. Finally, the user interface module for our system is reflected by a web app with a modern and engaging user interface and powered by Flask on the back-end. Fig. 2 provides a conceptual overview of our system, whereas Fig. 3 details the content processing and delivery pipeline.

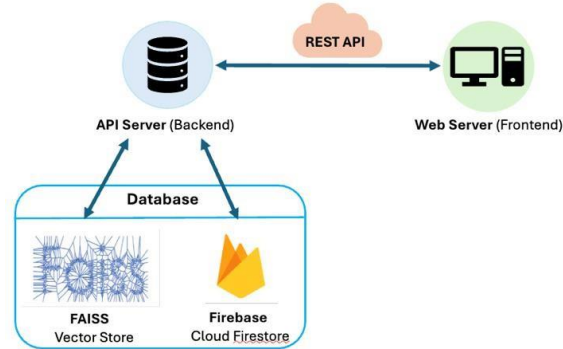


Fig. 2. An overview of our system's architecture.

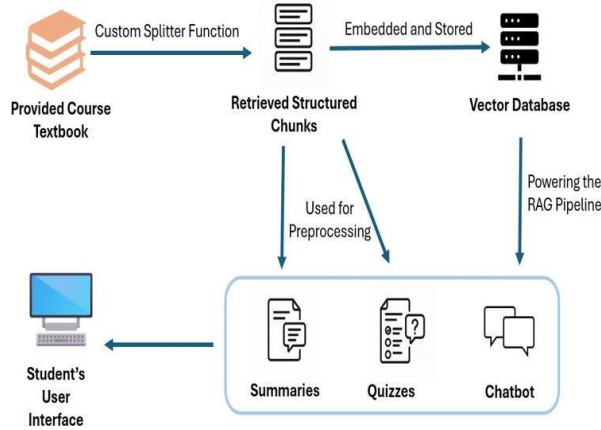


Fig. 3. General overview of our content processing and delivery pipeline.

The source textbook pdf sample we've decided to build our system on is [12], which consisted of over 40 chapters and 800 pages. With large amounts of text and context, LLMs and GenAI have shown significant weaknesses before [13], [14], often generating hallucinations and reducing factuality. Our system provides an effective solution to these problems and is composed of the following main components:

1. Summarized Lesson Modules and Formative Quizzes

To generate coherent, chapter and subchapter specific summaries from the Automata Theory textbook PDF [12], we first segment the document into a hierarchy of top-level chapters and nested subchapters. For efficient processing of textbook content, compression, and indexing strategies similar to those explored in [15] have been considered to optimize storage and retrieval performance. Each unit's page range is identified via our chapter & subchapter detection routine, and its text is extracted as an isolated corpus. These per-unit text blocks are then chunked and fed to our GenAI summarization pipeline, ensuring that each summary remains tightly focused on its intended section without contamination from unrelated material. Firstly, we parsed the pdf, and each line has been evaluated against the following carefully designed regular expressions (Regex):

Chapter Heading Regex:

$\wedge d+s+[A-Z][A-Za-z]+(?:s+[A-Za-z])+$

(matches lines beginning with a digit, followed by a space and a Title-Case word sequence)

Subchapter Heading Regex:

$\wedge d+\wedge d+s+[A-Z].*$

(matches any line that starts with a "digit.digit" section number (e.g 2.3) followed by one or more spaces and then a title beginning with an uppercase letter)

A sequential internal identifier (id) was then assigned to each identified chapter. The leading integer of the subchapter is then extracted and mapped to the corresponding chapter via the sequential id. The start page and the end page of each subchapter was also retrieved and stored in the database.

To then generate distilled conceptual summaries, we then reopened the textbook pdf and concatenated the text between each start and end page and submitted it to Gemini 2.0 Flash model with a specific prompt engineered to it. If the text in between is too large, we divide into more chunks, submit it to the LLM, and recombine the generated outcome. The outcome is then converted to HTML and formatted to appear in a neat readable format in the front-end. The system also makes sure to extract the diagrams for each subchapter by extracting their raw bytes directly from the PDF's XObject streams, base64-encoding them, then storing them in the database as data URIs.

The summaries of all the subchapters of each corresponding chapter are then all combined, retrieved, and sent to Gemini with a prompt engineered to ask it to output questions in multiple-choice format, prompting it to output a defined set of questions for each chapter, and for each question, it generates one correct answer, an array of incorrect answers, and a hint. The generated output is then saved under the Chapters collection in Firestore and displayed in the front-end, with a shuffling function applied to the multiple choices for each question.

Each layer, from PDF parsing through LLM-driven summarization and question generation, was carefully prompted and designed to be modular and resilient. Furthermore, upon successful login, students are first prompted to answer a simple questionnaire to check their knowledge background. Depending on the questionnaire's results, some summarized chapters become marked as skippable, and others become marked as recommended, offering a personalized learning path for each student.

2. RAG-Powered Interactive Chatbot

The PDF [12] is once again ingested, but this time its contents are divided into smaller yet still meaningful chunks. Breaking the knowledge-base backing a chatbot into smaller, uniformly-sized chunks has proved to yield better results not only for retrieval, but also for the accuracy of generated answers [16]. The chunks are then recursively split with a 1,000 character window, and a 100 character overlap, preserving semantic continuity across the chunk boundaries and outputting overlapping text segments optimized for embedding retrie

Each resulting text segment is then passed to the Google Generative AI Embeddings model [17], producing a dense, 1024-dimensional vector representation of the semantic content of the chunks. A FAISS index over the embedding vectors is then constructed, organizing the vectors into an approximate nearest-neighbor data structure and enabling sub-millisecond similarity queries at scale.

At inference time, the user's question is first embedded using the same model [17], and a k-nearest neighbor similarity search is then conducted, retrieving the four vectors whose dot-product with that query vector was the highest. These passages are then concatenated to form a context string, and together with the original user query, are supplied to Gemini LLM.

Grounding the generation in these retrieved excerpt constrains the output of our chatbot to verifiable content, therefore improving answer relevancy and factuality. Our current implementation has managed to achieve low-latency (<10 ms) retrieval for typical query loads and demonstrated strong alignment between generated explanations, and the source material, creating a chatbot solely dedicated to the course. The generation steps for our RAG pipeline are illustrated in Fig. 4.

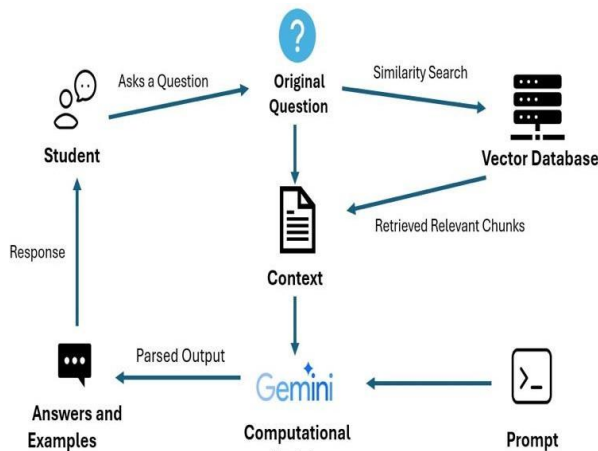


Fig. 4. Generation steps for the RAG pipeline of our chat interface.

3. Interactive Finite State Machine (FSM) Drawing Module

To not only offer passive knowledge consumption, but also active skill acquisition for a conceptually dense subject like Automata Theory, we have also developed an interactive FSM drawing module as shown in Fig. 5. This component represents a constructivist learning approach, providing students with a hands-on environment to apply theoretical concepts. Within the module, the system presents a student with a procedurally generated regular expression. The student then utilizes an intuitive, SVG-based canvas to visually construct a corresponding FSM, supporting actions such as creating and positioning states, defining transitions, and designating final states. This tool is designed to externalize the student's mental model, transforming an abstract problem-solving process into a concrete, interactive task that reinforces their understanding of how regular expressions relate to machine states and transitions.

The core innovation of this module is its AI-driven formative feedback loop, which is enabled by a novel serialization process. When a student chooses to "Check with AI," their visual, graphical FSM is algorithmically translated into a structured, textual description. This serialized representation formalizes the student's creation by listing its components—the set of states, alphabet, start state, final states, and a complete list of transitions. This text, along with the original regular expression, is then sent to a backend Gemini 2.0 Flash endpoint. The model evaluates the correctness of the student's FSM and returns immediate, actionable feedback. The response not only indicates whether the FSM is correct but also provides explanatory guidance for incorrect submissions, creating a powerful, iterative learning cycle where students can experiment, test their hypotheses, and refine their understanding.

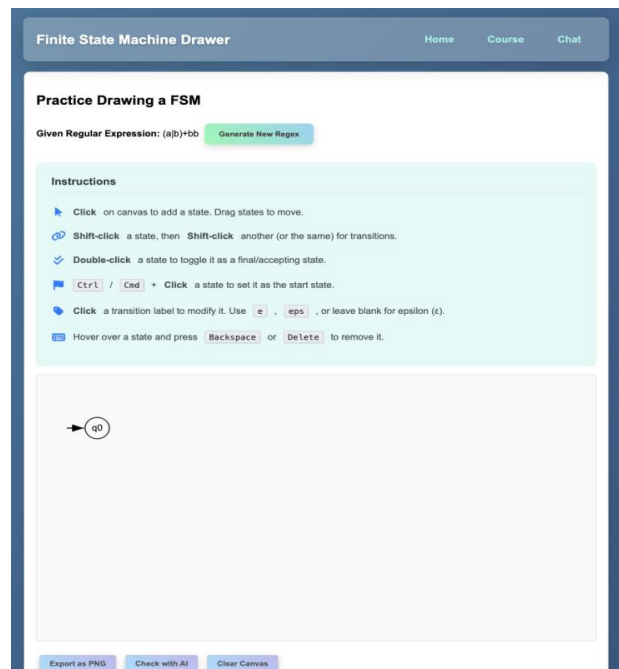


Fig. 5. Our system's interactive FSM drawing module for hands-on practice.

4. Personalized Progress Tracker

To give each student transparent insight of their studying journey, the ITS captures and aggregates two primary dimensions of engagement in reading activity and quiz performance. A dedicated dashboard (shown in Fig. 6) fetches and displays each student's tracked reading activity and quiz performance, offering the learners a concise image of their engagements and achievements. The dashboard updates on very minimal- payload API calls, allowing the system to scale efficiently under load. A lightweight front-end listener is also available, tracking and highlighting each chapter a user reads on the chapters list page. Together, these mechanisms give a concise, up to date view of their progress, prompting and motivating students to learn and engage more on the system.

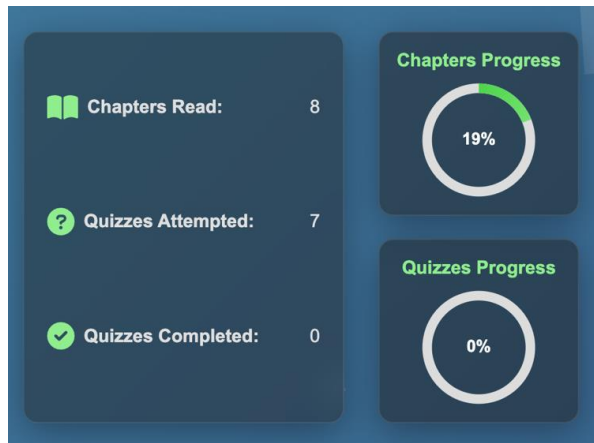


Fig. 6. Our system's personalized progress tracker.

IV. EVALUATION AND LIMITATIONS

To assess our system's effectiveness, we recruited 21 students who had previously completed the Automata Theory course available at our university to conduct a pilot study. Each participant engaged with the system by reviewing two distinct chapters, selected from the 42 available summarized modules, and completing the corresponding formative quizzes. Upon finishing these tasks, students completed a structured survey that included five-point Likert scales, ranging from 1 (hardly) and 5 (highly), to capture their subjective experiences. We have analyzed the students' responses based on their learning experience using our system, and their experience using some of our system's features. Table I. and Table II. detail our findings from the students' responses by analyzing the mean and standard deviations of their responses.

Overall, the responses were highly encouraging. On average, learners have highly agreed that the ITS increased their motivation to explore Automata Theory further (mean = 4.04/5) and reinforced their conceptual understanding of key topics (mean = 4.22/5). Participants have also depicted that they found elements of our system engaging and have shown a high tendency to recommend our system to other students learning Automata Theory. Furthermore, the students' feedback on the system's quizzes and chatbot were hugely

positive, indicating that our prompts and system pipeline are effective.

The Qualitative feedback we've received highlighted the system's strengths such as delivering focused, context-based explanations. These findings validate our design choices and illuminate promising directions for expanding our system's pedagogical impact. However, the students have also suggested a few future enhancements. For instance, a few of the students suggested an integration of video lessons and interactive simulations to complement text-based summaries, arguing that it would hugely benefit visual learners. Another suggested that having additional features such as flashcards would further increase the overall engagement of the learners.

It is important, however, to acknowledge the limitations of this pilot study. Firstly, the sample size of 21 students is small and may not be representative of a larger student population. Furthermore, the evaluation lacks a control group using traditional learning methods for comparison, making it difficult to attribute the positive outcomes solely to our ITS. Finally, our assessment relies on self-reported data (Likert scales) rather than objective measures of learning gain. A pre-test/post-test design would be necessary to quantify the actual improvement in students' knowledge. Therefore, these results should be interpreted as promising preliminary indicators that warrant a more comprehensive and controlled future evaluation.

TABLE I. SURVEY FINDINGS ANALYZING THE STUDENT'S LEARNING EXPERIENCE

Construct	Average (5-point Likert scale)	Standard Deviation
Increase in motivation to explore the subject	4.04	0.92
Conceptual Understanding Reinforcement	4.22	0.78
Engagement with Interactive Elements	4.0	0.80
Recommendation Likelihood	4.32	0.79

TABLE II. SURVEY FINDINGS ANALYZING SOME OF THE SYSTEM'S FEATURES

Construct	Average (5-point Likert scale)	Standard Deviation
Quiz Question Relevance	4.17	0.87
Chatbot Answer Relevance	4.40	0.89
Chatbot Responsiveness	4.28	0.87

V. SUMMARY AND FUTURE WORK

Through this paper, we have proposed and demonstrated a GenAI-powered ITS that makes a specialized contribution by integrating a suite of adaptive tools into a single, cohesive system for a formally rigorous course. Our work's novelty is centered on the synergy of its components: a hierarchical segmentation pipeline that overcomes the limitations of standard RAG for large textbooks; context-aware summaries and quizzes derived from this structure; and a fully integrated chatbot, all unified within one platform. This comprehensive architecture, applied to the challenging domain of Automata Theory, provides a solid framework for future ITS development. The system also features live progress tracking of each student's learning path, offering the students a comprehensive learning experience. Furthermore, our work builds on earlier advances in neural network performance optimization [18], [19] and web-based active learning frameworks [20], highlighting a natural evolution from classical e-learning platforms to AI-driven personalized tutoring systems.

In a pilot study involving 21 students who have previously completed the Automata Theory course, we observed encouraging preliminary results. To build upon these initial findings, our foremost priority for future research is to conduct a more rigorous evaluation. We aim to address the limitations of our evaluation by employing a larger student cohort and implementing a controlled experimental design. This will involve comparing an experimental group using our ITS against a control group using traditional study materials, such as textbooks or lecture notes. Additionally, to objectively measure learning outcomes, we plan on administering pre-test and post-test assessments to quantify knowledge gain directly.

Beyond a stronger evaluation, we plan on implementing additional features for visual learners, potentially using Python's Manim library to generate videos on the contents of our summaries. We also aim to add potential gamification and leaderboards. As according to [21], these elements have showed significant increases in engagement and interactions. Furthermore, we also aim to add a feature enabling teachers to monitor the students through the system itself. By combining these enhancements with a solid, evidence-based evaluation, we believe that we will have an enhanced well-rounded system for the students taking the Automata Theory course next semester at our university to use.

VI. REFERENCES

- [1] F. AlShaikh and N. Hewahi, "AI and Machine Learning Techniques in the Development of Intelligent Tutoring System: A Review," in *Proc. 2021 Int. Conf. on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT)*, Sakhir, Bahrain, Nov. 2021, pp. 403-410.
- [2] J. A. Kulik and J. D. Fletcher, "Effectiveness of intelligent tutoring systems: A meta-analytic review," *Review of Educational Research*, vol. 86, no. 1, pp. 42-78, 2016.
- [3] C. E. K. Agbewali-Koku, M. A. Rahman, M. Hamada, M. A. Ali, L. N. O. Oysharja, and M. T. Hossain, "A systematic review of machine learning techniques in online learning platforms," in *Proc. 2022 IEEE 15th Int. Symp. on Embedded Multicore/Many-core Systems (MCSoc)*, Penang, Malaysia, Dec. 19-22, 2022, pp. 247-250.
- [4] M. Hamada, "An Integrated Virtual Environment for Active and Collaborative e-Learning in Theory of Computation," *IEEE Trans. Learn. Technol.*, vol. 1, no. 2, pp. 117-130, Apr. 2008.
- [5] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535-547, Jul. 2019, doi: 10.1109/TBDATA.2019.2921572.
- [6] J. J. Slade, A. Hyk, and R. A. R. Gurung, "Transforming Learning: Assessing the Efficacy of a Retrieval-Augmented Generation System as a Tutor for Introductory Psychology," in *Proc. Human Factors and Ergonomics Society Annu. Meeting*, vol. 68, no. 1, Aug. 2024, pp. 1827-1830.
- [7] C. Q. Zhao, J. Cao, E. Chen, K. R. Koedinger, and J. Lin, "SlideItRight: Using AI to Find Relevant Slides and Provide Feedback for Open-Ended Questions," *arXiv:2505.04584*, May 2025.
- [8] Z. Ye, "Intelligent Tutoring Agent with Retrieval-Augmented Generation: A Case Study of Quality Management System Course," in *Proc. 2025 5th International Conference on Consumer Electronics and Computer Engineering (ICCECE)*, Ningbo, China, 2025, pp. 189-194.
- [9] P. Bassner, E. Frankford, and S. Krusche, "Iris: An AI-Driven Virtual Tutor for Computer Science Education," in *Proc. Innovation and Technology in Computer Science Education (ITI-CSE 2024)*, vol. 1, Jul. 2024, pp. 1-7.
- [10] R. W. C. Lui, H. Bai, A. W. Y. Zhang, and E. T. H. Chu, "GPTutor: A Generative AI-powered Intelligent Tutoring System to Support Interactive Learning with Knowledge-Grounded Question Answering," in *Proc. 2024 Int. Conf. on Advances in Electrical Engineering and Computer Applications (AEECA)*, Dalian, China, Aug. 16-18, 2024, pp. 702-707.
- [11] E. Mousavinasab, N. Zarifsanaiy, S. R. Niakan Kalhori, M. Rakhshan, L. Keikha, et al., "Intelligent tutoring systems: A systematic review of characteristics, applications, and evaluation methods," *Interactive learning environments*, vol. 29, no. 1, pp. 142-163, 2021.
- [12] E. Rich, *Automata, Computability, and Complexity: Theory and Applications*, 1st ed. Upper Saddle River, NJ, USA: Pearson Education, 2008.
- [13] R. Azamfirei, S. R. Kudchadkar, J. Fackler, "Large language models and the perils of their hallucinations," *Critical Care* 27 (2023) 120.
- [14] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell, P. Welinder, P. F. Christiano, J. Leike, R. Lowe, "Training language models to follow instructions with human feedback," in: *NeurIPS*, 2022.
- [15] M. A. Rahman and M. Hamada, "Burrows-wheeler transform based lossless text compression using keys and Huffman coding," *Symmetry*, vol. 12, no. 10, p. 1654, 2020.
- [16] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense Passage Retrieval for Open-Domain Question Answering," in *Proc. 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Virtual, Nov. 16-20, 2020, pp. 6769-6781.
- [17] Google, "Google Generative AI Embeddings API," 2024. [Online]. Available: <https://developers.google.com/resources/embeddings>.
- [18] M. Hassan and M. Hamada, "Performance comparison of feed-forward neural networks trained with different learning algorithms for recommender systems," *Computation*, vol. 5, no. 3, p. 40, 2017.
- [19] H. M. El-Bakry and M. Hamada, "A new implementation for high-speed normalized neural networks in frequency space," *Lect. Notes Comput. Sci.*, vol. 5177, pp. 33-40, 2008.
- [20] M. Hamada, "Web-based tools for active learning in information theory," in *Proc. SIGCSE Tech. Symp. Comput. Sci. Educ.*, 2007, pp. 60-64.
- [21] H. Çiğdem, M. Öztürk, Y. Karabacak, N. Atik, S. Gürkan, and M. H. Aldemir, "Unlocking student engagement and achievement: The impact of leaderboard gamification in online formative assessment for engineering education," *Education and Information Technologies*, vol. 29, pp. 24835-24860, Jun. 2024, doi:10.1007/s10639-024-12845-2.